

Aberrant FGF signaling promotes granule neuron precursor expansion in SHH subgroup infantile medulloblastoma

Reviewed Preprint

v1 • September 12, 2024

Not revised

Odessa R Yabut , Hector Gomez, Jessica Arela, Jesse Garcia Castillo, Thomas Ngo, Samuel J Pleasure 

Department of Neurology, Weill Institute for Neuroscience, University of California San Francisco, San Francisco, California • Department of Psychiatry, Weill Institute for Neuroscience, University of California San Francisco, San Francisco, California, USA • Programs in Neuroscience and Developmental Biology, Eli and Edythe Broad Center of Regeneration Medicine and Stem Cell Research, University of California San Francisco, California, USA

 https://en.wikipedia.org/wiki/Open_access
 Copyright information

Abstract

Mutations in Sonic Hedgehog (SHH) signaling pathway genes, e.g. *Suppressor of Fused* (SUFU), drive granule neuron precursors (GNP) to form medulloblastomas (MB^{SHH}). However, how different molecular lesions in the Shh pathway drive transformation is frequently unclear, and SUFU mutations in the cerebellum seem distinct. In this study, we show that fibroblast growth factor 5 (FGF5) signaling is integral for many infantile MB^{SHH} cases and that FGF5 expression is uniquely upregulated in infantile MB^{SHH} tumors. Similarly, mice lacking SUFU (Sufu-cKO), ectopically express FGF5 specifically along the secondary fissure where GNPs harbor preneoplastic lesions and show that FGFR signaling is also ectopically activated in this region. Treatment with an FGFR antagonist rescues the severe GNP hyperplasia and restores cerebellar architecture. Thus, direct inhibition of FGF signaling may be a promising and novel therapeutic candidate for infantile MB^{SHH}.

eLife assessment

This study provides potentially highly **valuable** new insight into the role of Fgf signalling in SUFU mutation-linked cerebellar tumors and indicates novel therapeutic interventions via inhibition of Fgf signalling. The evidence supporting the major claims, however, is at this point currently **incomplete**. A more robust analysis of gene expression patterns and deeper mechanistic insight would significantly enhance this study, which could have wide-ranging implications for the treatment of specific cerebellar tumors.

<https://doi.org/10.7554/eLife.100767.1.sa4>

Introduction

Medulloblastoma (MB) is the most common malignant brain tumor in children, with half of cases diagnosed before the age of 5 (Ward et al., 2014; Ostrom et al., 2016). Mutations in *Suppressor of Fused (SUFU)* comprise approximately 30% of tumors in infants with Sonic Hedgehog driven MB (MB^{SHH}). Infantile MB^{SHH}, including those with SUFU mutations (MB^{SHH-SUFU}) have worse prognosis and higher rates of local recurrence than other MB^{SHH} subtypes (Kool et al., 2014; Schwalbe et al., 2017; Guerrini-Rousseau et al., 2018). Unfortunately, available SHH-targeted treatments for MB^{SHH} act specifically on proteins upstream of SUFU and are therefore ineffective for MB^{SHH-SUFU} patients (Kool et al., 2014). The poor prognosis, early occurrence, and lack of targeted therapy for MB^{SHH-SUFU} patients make detailed understanding of the drivers of oncogenesis in this group of great importance.

Sufu acts as an intracellular modulator of SHH signaling (Matise and Wang, 2011). Briefly, the SHH signaling pathway is initiated after binding of extracellular SHH ligands to the transmembrane receptor Patched 1 (Ptch1). This relieves Ptch1 inhibition of the transmembrane protein, Smoothened (Smo), and enables the initiation of a cascade of intracellular events promoting the activator function of the transcription factors, Gli1, Gli2, or Gli3. Sufu modulates SHH signaling by ensuring the stability of Gli transcription factors or by promoting the formation of the repressor forms of Gli2 (Gli2R) or Gli3 (Gli3R) (Chen et al., 2009; Wang et al., 2010; Lin et al., 2014). Thus, depending on the developmental context, loss of Sufu can lead to activation or repression of SHH signaling. In the developing cerebellum, Sufu dysfunction is associated with abnormal development of granule neuron precursors (GNP), which account for MB^{SHH} (Kim et al., 2011, 2018; Vanner et al., 2014; Kong et al., 2019; Vladoiu et al., 2019; Yin et al., 2019; Jiwani et al., 2020). GNPs populate the external granule layer (EGL) along the cerebellar perimeter where local SHH signals trigger GNP proliferation and differentiation at neonatal stages. However, other mitogenic pathways can also influence GNP behavior (Leto et al., 2016), yet little is known on how Sufu interacts with these pathways. Understanding how SUFU loss-of-function (LOF) affects the activity of concurrent local signaling pathways in granule neuron development may be key towards developing potent targets for anti-tumor therapy.

In this study, we examined regulation of FGF signaling in MB^{SHH} and identified upregulation of FGF5 expression in tumors of infantile MB^{SHH} patients. Similarly, we show ectopic expression of FGF5 in the neonatal cerebellum of mice lacking SUFU and correlates with the activation of FGF signaling in surrounding EGL-localized cells where GNPs accumulate. Strikingly, acute pharmacological inhibition of FGF signaling results in near-complete rescue of these defects, including a restoration of cerebellar histo-organization. Thus, our findings identify FGF5 as a potential biomarker for a subset of patients with infantile MB^{SHH} who may be responsive to FGFR-targeting therapies.

Results

FGF5 is specifically upregulated in SHH-driven infantile MB

We previously reported that SUFU LOF in neocortical progenitors result in FGF signaling activation to influence specification and differentiation of neocortical excitatory neurons (Yabut et al., 2020). Thus, we sought to determine if key FGF signaling pathway genes are differentially expressed in MB patient tumors. We performed comparative analysis of the expression dataset from 763 MB patient samples comprised of tumors resected from molecularly distinct MB subgroups: Wingless (Wnt subgroup; MB^{WNT}), MB^{SHH}, Group 3 (MB^{Group3}), and Group 4

(MB^{Group4}) (Cavalli et al., 2017 [↗](#)). Strikingly, our analyses show that FGF5 expression is higher in tumors specifically from MB^{SHH} patients compared to other MB subgroups, with approximately 25% of MB^{SHH} tumors exhibiting a two-fold increase (**Fig. 1A-1B** [↗](#)). We also find that FGF5 is uniquely upregulated in MB^{SHH} tumors from patients within the 0-3 years old age group, but not patients within the same age group in other MB subtypes (**Fig. 1C** [↗](#)). Further examination across all MB^{SHH} tumors stratified which subgroups express the highest levels of FGF5 expression. Infantile tumors, largely belonging within the SHHb and SHHg subgroups (Cavalli et al., 2017 [↗](#)), exhibit higher FGF5 levels compared to tumors from children (SHHa) or adults (SHHd) (**Fig. 1D** [↗](#)). By all measures, the proportion of SHHb and SHHg tumors with relatively high levels of FGF5 is significantly increased (~30%) compared to other MB^{SHH} subgroups (**Fig. 1E** [↗](#)). Taken together, these findings strongly suggest that FGF signaling is specifically disrupted in infantile-onset MB^{SHH}.

Region-specific expansion of GNPs in the P0 Sufu-cKO cerebellum coincides with increased FGF5 expression

FGF signaling has been implicated in cerebellar development, particularly in granule neuron development (Yaguchi et al., 2009 [↗](#); Yu et al., 2011) leading us to wonder if and how aberrant FGF signaling may be contributing to oncogenicity of GNPs. Since mutations in *SUFU* drive infantile MB^{SHH}, we generated the mutant mice (*hGFAP-Cre;Sufu^{fl/fl}*, hereto referred as Sufu-cKO), in which Sufu is conditionally deleted in GNPs (Zhuo et al., 2001 [↗](#)) to examine FGF expression and activity. Sufu-cKO mice exhibit profound defects in cerebellar development. At P0, a timepoint at which GNP proliferation and differentiation is ongoing, there is a visible increase in measured cerebellar size and expansion of Pax6-positive (Pax6+) GNPs in the EGL of Sufu-cKO cerebellum compared to controls (**Fig. 2A-C** [↗](#)). Notably, the expansion of Pax6+ GNPs specifically localize along the secondary fissure (EGL Region B, arrow in **Fig. 2A** [↗](#)) compared to other EGL areas (EGL Regions A and C) in the P0 Sufu-cKO cerebellum (**Fig. 2D** [↗](#)). We then proceeded to examine whether these defects correlated with abnormal FGF5 expression. *In situ* hybridization (ISH) using FGF5-specific riboprobes show high expression of FGF5 (FGF5^{high}) immediately adjacent to the secondary fissure in the P0 control cerebellum (**Fig. 2E** [↗](#)). Strikingly, in the P0 Sufu-cKO cerebellum, there is anterior expansion of FGF5^{high} expression regions (outlines, **Fig. 2F** [↗](#)), coinciding with areas near the secondary fissure where GNP expansion is most severe (**Fig. 2A, 2D** [↗](#)). Further, while FGF5-expressing (FGF5+) cells are largely excluded from the EGL of the control cerebellum, a substantial number of FGF5+ cells are ectopically localized in the EGL Region B of the Sufu-cKO cerebellum (**Fig. 2E** [↗](#)). FGF5 mRNA molecules are visibly higher in FGF5-expressing cells in the Sufu-cKO EGL compared to controls (arrows, **Fig. 2F** [↗](#)). These findings implicate FGF5 as a potential instigator of region-specific defects in GNP differentiation present in the P0 Sufu-cKO cerebellum.

FGF signaling drives GNP proliferation in the P0 Sufu-cKO cerebellum

FGF5 is a ligand for fibroblast growth factor receptors 1 (FGFR1) and 2 (FGFR2), both of which are expressed in the developing cerebellum, particularly in IGL regions where FGF5-expressing cells localize (Clements et al., 1993 [↗](#); Ornitz et al., 1996 [↗](#)). Binding of FGF5 to these receptors triggers the activation of multiple intracellular signaling pathways including the mitogen activated protein kinase (MAPK) pathway to control cellular activities driving NSC progression (**Fig. 3A** [↗](#)) (Ornitz and Itoh, 2015 [↗](#)). Immunostaining with antibodies to detect MAPK pathway activation reveals increased MAPK activity in the EGL of the P0 Sufu-cKO cerebellum. Regional distribution of cells labeled with phospho-Erk1/2 (pErk1/2+), a marker for activated MAPK signaling, shows that the abnormally expanded EGL of the secondary fissure in the P0 Sufu-cKO cerebellum increase in these cells (**Fig. 3B** [↗](#)). Quantification of the number of pERK1/2+ cells in this area reveals that this is significant compared to controls (**Fig. 3C** [↗](#)). Many pERK1/2+ cells are also proliferative as indicated by Ki67 labeling (boxed areas, **Fig. 3B** [↗](#)) and the numbers of these dual labeled cells are

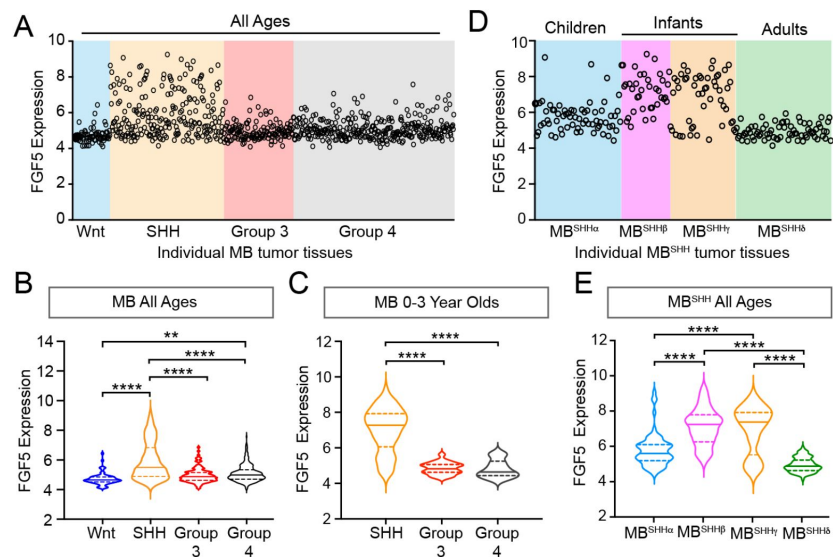


Figure 1

Upregulated FGF5 expression in MB_{SHH} tumors from infant patients.

(A) Levels of FGF5 expression in human MB tumors of all ages from GEO expression dataset #GSE85217 (Cavalli et al., 2017). (B, C) Statistical analysis of FGF5 expression levels associated with MB tumor subtypes from patients across all ages (B) and 0-3 year old MB patients (C). ** $p < 0.01$, **** $p < 0.0001$. (D, E) Graph represents FGF5 expression levels in human MB_{SHH} tumors of all ages from GEO expression dataset #GSE85217 (D) and corresponding plots (E) showing statistically higher FGF5 expression in tumors from infants with MB_{SHH} compared to tumors from children or adults with MB_{SHH}. **** $p < 0.0001$.

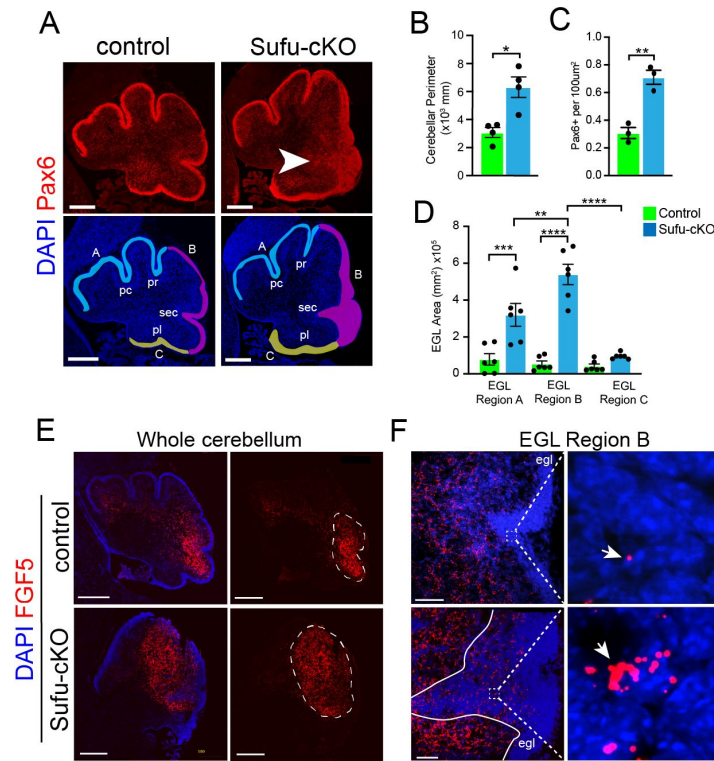


Figure 2

Increased FGF5 expression coincides with region-specific expansion of GNPs in the P0 Sufu-cKO cerebellum.

(A) Pax6 (red) and DAPI (blue) immunofluorescence staining of the P0 Sufu-cKO and control cerebelli. Arrow points to severely expanded EGL region B in the P0 Sufu-cKO cerebellum. EGL regions are designated in DAPI-labeled sections as A (light blue), B (magenta), and C (yellow). Each region encompasses specific fissures: the preculminate (pc) and primary (pr) fissures for Region A, the secondary (sec) fissure for Region B, and the posterolateral (pl) fissure for Region C. Scale bars: Scale bars = 250 µm. **(B-D)** Quantification and comparison of the cerebellar perimeter (B), total area occupied by densely packed Pax6+ cells (C), and size of specific EGL regions (D) between P0 Sufu-cKO and control cerebelli. **(E)** Fluorescent *in situ* hybridization using RNAScope probes against FGF5 mRNA (red) and DAPI labeling in the P0 Sufu-cKO and control whole cerebellum. Areas with high levels of FGF5 expression are outlined. Scale bars = 500 µm. **(F)** FGF5, as detected by fluorescent *in situ* hybridization, is ectopically expressed by cells in the EGL and adjacent regions of Region B. Boxed areas within the EGL show DAPI-labeled cells expressing visibly high levels of FGF5 (arrowheads) in the EGL of P0 Sufu-cKO cerebellum compared to controls. Scale bars = 50 µm. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

significantly higher in the Sufu-cKO cerebellum compared to controls (**Fig. 3D**). These findings indicate that the increase in FGF5 correlates with the activation of FGF signaling in GNPs and demonstrates a likely role in regulating the abnormal proliferation and pre-neoplastic lesion in mutant mice.

Given the role of SUFU in regulating Gli transcription factors to modulate SHH signaling activity (Kim et al., 2018; Yin et al., 2019), we examined whether the activation of FGF signaling occurred concurrently with SHH signaling activation. Surprisingly, Gli protein levels in the P0 control and Sufu-cKO cerebellum show marked reduction of Gli1, Gli2, Gli3, and Ptch1 levels (**Supplementary Fig. 1A**); this is inconsistent with elevated SHH signaling we anticipated in the absence of SUFU. To directly examine this in specific EGL regions, we compared the cerebellum of P0 control and Sufu-cKO mice carrying the SHH reporter transgene Gli1-LacZ (Ahn and Joyner, 2005). In these mice, SHH signaling activity is absent or very low throughout the entire P0 Sufu-cKO cerebellum but is highly active in a region-specific manner in controls (**Supplementary Fig. 1B**). Furthermore, while some LacZ⁺ cells are detectable in EGL regions A and C and adjacent ML and IGL, LacZ⁺ cells are completely absent in EGL region B and adjacent areas of the P0 Sufu-cKO cerebellum (**Supplementary Fig. 1B-1C**). These findings indicate that the accumulation of GNPs does not rely on active SHH signaling, particularly in Region B where there is a severe expansion of GNPs in the absence of SUFU.

Blockade of FGF signaling dramatically rescues the Sufu-cKO phenotype

To determine if activated FGF signaling drives GNP defects in the Sufu-cKO cerebellum, we pharmacologically inhibited FGF signaling using the competitive FGFR1-3 antagonist AZD4547 (Gudernova et al., 2016). For this experiment, 1 µl of AZD4547 (5mg/ml) was delivered via intraventricular (IV) injection for 5 consecutive days beginning at P0 and the cerebellum analyzed 3 days after treatment at P7 (**Fig. 3E**). Strikingly, AZD4547 treatment results in near complete rescue of the GNP phenotype by P7 in the Sufu-cKO cerebellum, displaying a cerebellar morphology indistinguishable from controls with normal foliation and cellular organization (**Fig. 3F**). Indeed, in the cerebellum of AZD4547-treated P7 Sufu-cKO mice, proliferating Ki67⁺ cells largely exclusively localize in the EGL while NeuN⁺ cells densely pack the IGL and not the EGL (**Fig. 3G**). Notably, NeuN expression appears in cells localized at the border of the EGL and ML, where Ki67⁺ cells are absent, indicating that post-mitotic cells successfully began differentiation as observed in controls (boxed regions, **Fig. 3G**). Thus, our findings confirm that inhibition of FGF signaling in proliferating GNPs of the Sufu-cKO cerebellum ensure normal progression of GNP differentiation.

FGF5 expression is increased in the developing cerebellum of Sufu;p53-dKO mice

Our findings indicate a critical role for FGF signaling in driving GNP hyperplasia, making GNPs vulnerable to neoplastic lesions resulting in tumorigenesis, when SUFU is absent in the developing cerebellum. Indeed, we find that Pax6⁺ GNPs within the neonatal Sufu-cKO EGL display an increase in double-strand breaks, especially in Region B (DSBs; **Fig. 4A-4B**), as detected by immunostaining for phosphorylated H2AX (γH2AX), an early marker for DSBs (Mah et al., 2010). Nevertheless, as previously reported, tumors do not readily form in the Sufu-cKO cerebellum (Yin et al., 2019), indicating either timely repair of DSBs or the induction of apoptosis in GNPs with significant genomic instability. Indeed, double-labeling with cleaved Caspase 3 (CC3) and γH2AX show a significantly higher number of double-labeled cells in EGL Regions A and B (**Supplementary Fig. 2A-2B**). Among the downstream targets of DSB repair pathways is p53, which when activated mediates cell death to suppress tumor formation (Gao et al., 2000). In the P0 Sufu-cKO cerebellum, p53 protein is present, albeit significantly reduced (**Fig. 4C**). The reduction in p53 may be driving an increase in DSBs yet still sufficient to induce apoptotic

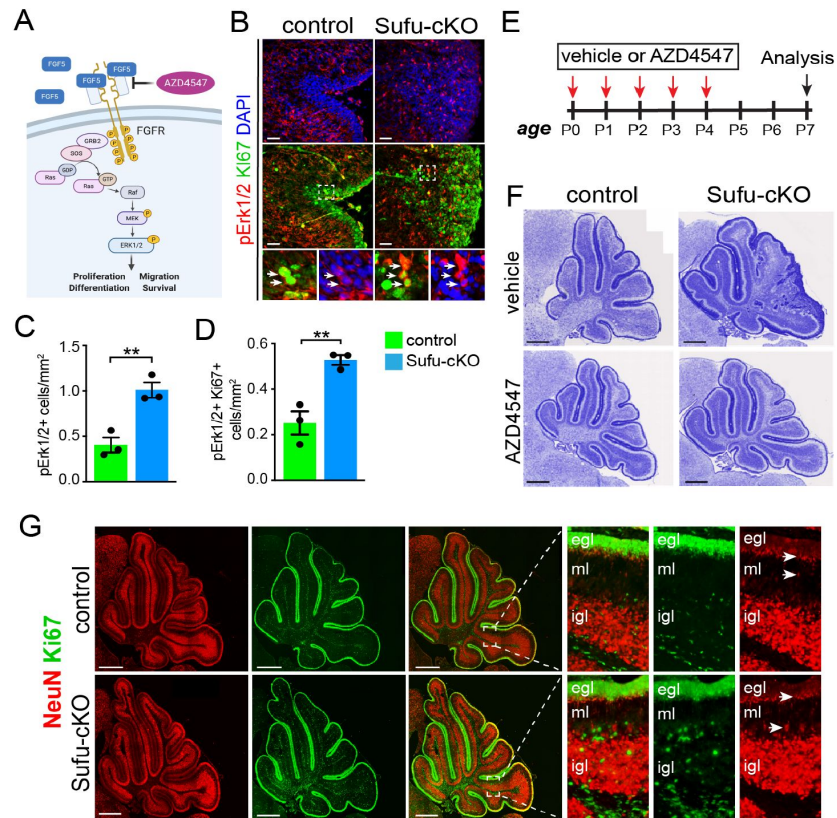


Figure 3

Ectopic activation of FGF signaling in the EGL of P0 Sufu-cKO cerebellum.

(A) Schematic diagram showing the activation of FGF signaling activity, upon binding of FGF5 to extracellular domains of FGFR, via the MAPK signal transduction pathway. **(B)** Double-immunofluorescence staining with Ki67 (green) and phospho-Erk1/2 (pErk1/2; red), a marker of activated MAPK signaling in the P0 Sufu-cKO and control cerebelli. Boxed regions show pErk1/2+ and Ki67+ cells (arrowheads) in the control and Sufu-cKO EGL. Scale bars = 50 μ m. **(C, D)** Quantification of pErk1/2+ cells (C) and double-labeled pErk1/2+ and Ki67+ cells (D) in the P0 Sufu-cKO and control EGL Region B. ** $p < 0.01$. **(E)** Experimental design of rescue studies performed by intraventricular administration of FGFR1-3 pharmacological inhibitor, AZD4547 or vehicle controls. **(F)** Nissl staining of the P7 control and Sufu-cKO treated with either AZD4547 or vehicle, 2 days after treatment. Scale bars = 500 μ m. **(G)** NeuN and Ki67 double immunofluorescence staining of the P7 control and Sufu-cKO treated with either AZD4547 or vehicle. Boxed regions show localization and organization of NeuN+ and Ki67+ cells in distinct cerebellar layers. Arrows point to areas of the EGL and IGL where NeuN+ cells are beginning to be expressed.

pathways. Supporting this, conditional ablation of both p53 and SUFU in GNPs (*hGFAP-Cre;Sufu^{fl/fl};p53^{fl/fl}* or *Sufu;p53-dKO*) results in the formation of massive tumors in the cerebellum within 2 months after birth (**Fig. 4F**), indicating the failure to activate critical apoptotic pathways. However, tumors do not form in mice lacking p53 (Marino et al., 2000) or SUFU alone (**Fig. 4F**). These findings suggest that in the absence of SUFU, the failure of GNPs to transition into fully differentiated granule neurons compromises genomic stability and renders GNPs extremely vulnerable to tumor formation with a second molecular hit.

We sought to confirm that upregulated FGF signaling also occur in the tumor-prone *Sufu;p53-dKO*. As expected, the neonatal cerebellum of *Sufu-cKO*, display EGL expansion due to excess proliferative (Ki67+) and Pax6+ cells in the P0 *Sufu;p53-dKO* cerebellum but not in *p53-cKO* and control cerebelli (**Fig. 4G**). Further expansion of Pax6+ GNPs in the P0 *Sufu;p53-dKO* is also most severe along the secondary fissure (EGL Region B) compared to other EGL areas (EGL Regions A and C) of the P0 *Sufu;p53-dKO* cerebellum (**Fig. 4G-4H**). As with our observations in the P0 *Sufu-cKO* cerebellum, ISH for FGF5 in the P0 *Sufu;p53-dKO* cerebellum show ectopic FGF5 expression. Particularly, FGF5+ cells are expanded anteriorly and detected specifically around the secondary fissure of the P0 *Sufu;p53-dKO* cerebellum (**Fig. 4I**). There is also ectopic MAPK signaling activity in the P0 *Sufu;p53-dKO* cerebellum, with significantly higher numbers of pErk1/2+ cells within the EGL, many of which are proliferative as marked by co-labeling with Ki-67, within the expanded EGL of the secondary fissure (**Fig. 4J**). These findings indicate that as in the P0 *Sufu-cKO* cerebellum, ectopic FGF5 expression triggers FGF signaling in GNPs in the P0 *Sufu;p53-dKO* cerebellum and may facilitate oncogenic transformation and tumor growth of GNPs.

Discussion

Our studies identify a mechanism by which the combinatorial effects of oncogenic *SUFU* mutations and other concurrent developmental signaling pathways, make GNPs vulnerable to oncogenic transformation leading to infantile MB^{SHH}. Using mice lacking SUFU in GNPs, we find that ectopic FGF5 expression correlates with an increase in FGF signaling, particularly in areas where proliferating GNPs reside, resulting in GNP hyperplasia, preneoplastic lesions, and patterning defects. Inhibition of FGF signaling through pharmacological blockade of FGFR1-3 prevents hyperplasia and associated cerebellar architectural abnormalities. Strongly supporting a role for FGF5, we also find elevated levels of FGF5 gene expression specifically in infantile MB^{SHH} patients, but not in other MB subgroups. These findings indicate that FGF-targeting compounds may be a promising therapeutic option for infantile MB^{SHH} patients, with elevated levels of FGF5 in tumor tissues acting as a potential biomarker.

Expansion of Pax6+ GNPs in the newborn *Sufu-cKO* cerebellum (**Fig. 2**) occur in similar anatomic subregions where infantile MB tumors typically arise (Tan et al., 2018). Interestingly, these subregions have low levels of Gli1 reporter activity, Ptch1 expression, and SHH ligand expression (Corrales et al., 2004). Spatially distinct regulation of granule neuron development by *Sufu* may therefore rely on non-canonical SHH signaling beyond *Smo*, or through yet undefined downstream interactions resulting in control of FGF signaling activity. Supporting this, we find ectopic expression of FGF5 and deregulation of FGF signaling as marked by the excessive intracellular activation of MAPK signaling in the P0 *Sufu-cKO* cerebellum. Similarly in previous studies, ectopic expression of FGF ligands, FGF8 and FGF15, are associated with degradation of Gli3R because of SUFU deletion, resulting in regional patterning and precursor specification and differentiation defects (Kim et al., 2011, 2018; Jiواني et al., 2020; Yabut et al., 2020). Taken together, these findings indicate that SUFU acts at the intersection of SHH and FGF signaling to modulate GNP behavior (as summarized in our working model in **Fig. 8**). We postulate that SUFU exerts this role via stabilization of Gli transcription factors, since we observed a decrease in Gli1, Gli2, and Gli3 protein level in the newborn *Sufu-cKO* cerebellum. Supporting this, Yin et al.,

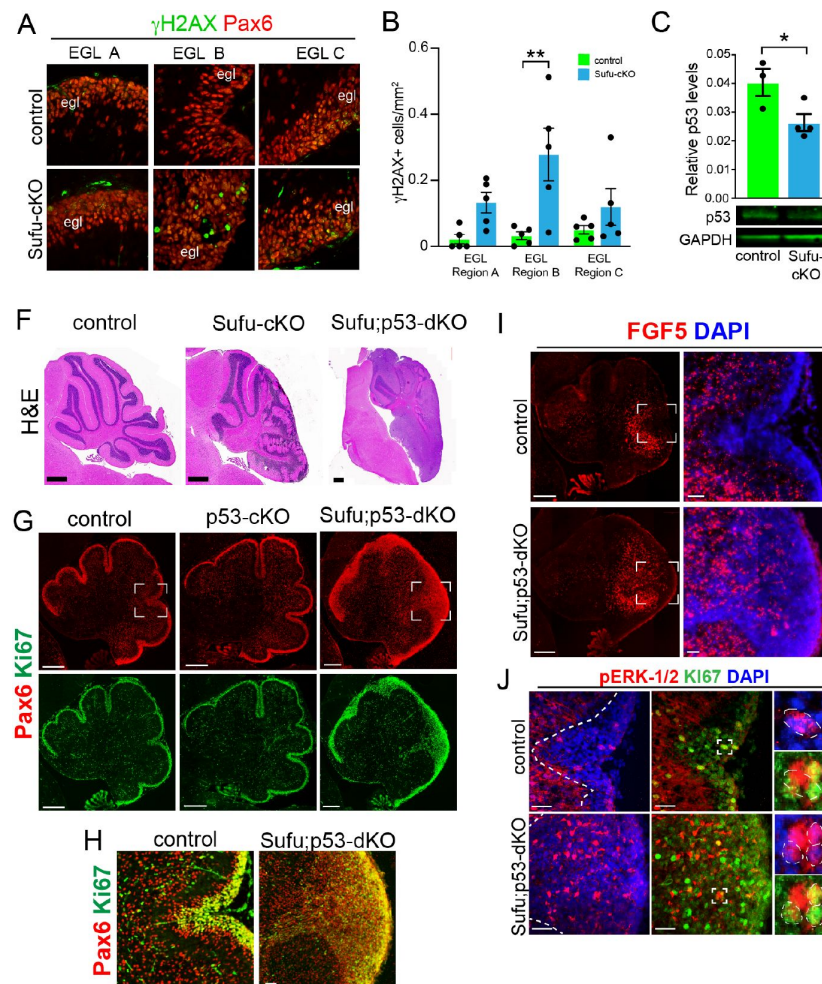


Figure 4

Evidence of pre-neoplastic lesions and high rates of cell death in Sufu-cKO granule neuron precursors.

(A) Double-immunofluorescence staining with Pax6 (red) and γ H2AX (green), a marker for double-strand DNA breaks in specific EGL regions of the P0 Sufu-cKO and control cerebelli. (B) Quantification of γ H2AX+ cells in each cerebellar region of P0 control and Sufu-cKO mice. ** $p < 0.01$. (C) Western blot analysis of p53 protein levels in P0 control and Sufu-cKO cerebellar protein lysates. * $p < 0.05$. (F) Hematoxylin and Eosin (H&E) staining of P60 control, Sufu-cKO, Sufu;p53-dKO cerebelli. Scale bars = 500 μ m. (G, H) Double-immunofluorescence staining against Pax6 (red) and Ki67 (green) in the P0 control, p53-cKO, and Sufu;p53-dKO cerebellum (G). Boxed regions in G are magnified in H demonstrating the expansion of the EGL in the P0 Sufu;p53-dKO cerebellum compared to littermate controls. Scale bars = 200 μ m (A) and 50 μ m (B). (I) Fluorescent *in situ* hybridization using RNAScope probes against FGF5 mRNA (red) and DAPI labeling in the P0 Sufu;p53-dKO and control cerebellum. Boxed areas are enlarged to show ectopic localization of FGF5+ cells in the EGL of Sufu;p53-dKO cerebellum unlike in controls. Scale bars = 200 μ m and 50 μ m (boxed area). (J) Double-immunofluorescence staining with Ki67 (green) and phospho-Erk1/2 (pErk1/2; red) in the P0 Sufu;p53-dKO and control cerebelli. Boxed regions show cells double-labeled with pErk1/2+ and Ki67+ cells in the control and Sufu;p53-dKO EGL Region B. Scale bars = 25 μ m.

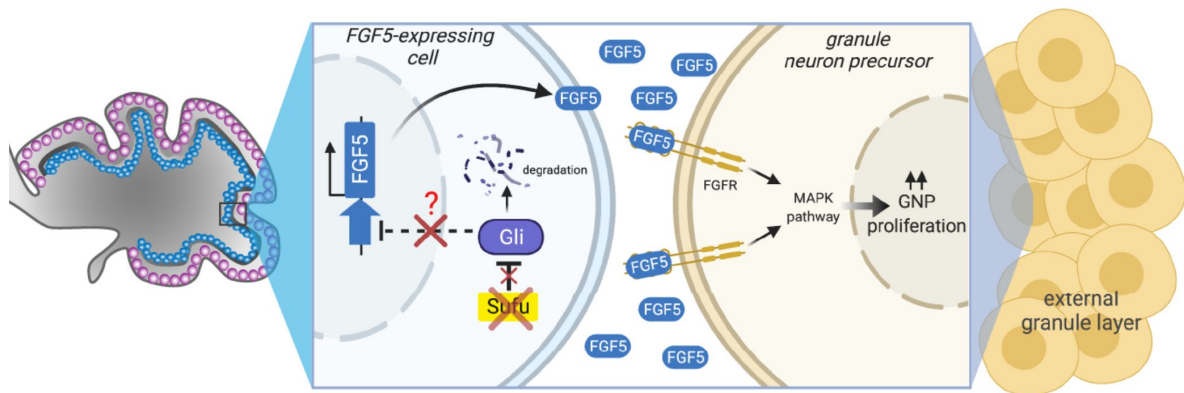


Figure 5

Loss of Sufu function drives excess proliferation of granule neuron precursors via FGF signaling activation.

The schematic diagram summarizes how Sufu LOF facilitate expansion of GNPs at early stages of cerebellar development by driving FGF signaling activation via ectopic expression of FGF5.

(2019) [found](#) that reducing Gli2 levels rescued GNP hyperplasia and patterning defects in the Sufu-cKO neonatal cerebellum. Further studies are required to elucidate the involvement of Gli-dependent mechanisms in controlling FGF5 gene expression in the neonatal cerebellum.

In the wildtype cerebellum, FGF5-expressing cells localize within the IGL surrounding the secondary fissure at P0 (**Fig. 4B**) (Ozawa et al., 1996; Hattori et al., 1997; Yaguchi et al., 2009) where Gli1 reporter activity (**Fig. 4A**), and SHH and Ptch1 expression are lowest (Corrales et al., 2004). By P4, FGF5 expression is detected in cells within the IGL throughout the cerebellum (Yaguchi et al., 2009). However, by P14, a timepoint at which most granule neurons have differentiated, FGF5 is no longer detected in the IGL or other cerebellar regions per Allen Developing Mouse Brain Atlas (www.brain-map.org). The strict spatiotemporal expression of FGF5 strongly supports a stage-specific role for regulating GNP development, particularly along the secondary fissure, to ensure timely differentiation and maturation of granule neurons. Indeed, deregulation of FGF5 expression, as we observed in the P0 Sufu-cKO cerebellum, results in the inability of adjacent GNPs to normally differentiate and instead continued massive proliferation.

Our findings are contrary to previous reports of the proliferation-suppressive roles of FGFs, particularly FGF2, in GNPs specifically carrying *Ptch1* mutations (Fogarty et al., 2007; Emmenegger et al., 2013). However, in contrast to the *Ptch1*-cKO cerebellum, the newborn Sufu-cKO cerebellum still express *Ptch1* and exhibit reduced Gli1, Gli2, and Gli3 protein levels (**Supplementary Fig. 1A**). These key molecular differences may activate unique signaling networks in Sufu-cKO GNPs. Additionally, there are 18 FGF ligands which differ significantly in molecular features and binding specificities to distinct combinations of FGFR splice variants (Ornitz and Itoh, 2015). For example, unlike FGF2 which act in an autocrine manner, FGF5 can exert both autocrine and paracrine functions, bind different combinations of FGFRs, and is dynamically expressed in distinct regions of the developing cerebellum. Thus, in-depth studies are needed to elucidate the exact mechanisms triggered by abnormally high levels of FGF5 in the developing cerebellum, particularly since FGF5 overexpression is known to drive cancer development and progression, including brain tumors (Allerstorfer et al., 2008).

The high occurrence of *SUFU* mutations in infantile MB indicate the selective vulnerability of the developing cerebellum to neoplastic effects of *SUFU* dysfunction. Notably, the timing of *SUFU* LOF is critical; conditional deletion of *SUFU* in neural stem cells prior to granule neuron specification (using the hGFAP-Cre line) results in GNP hyperplasia, whereas conditional deletion of *SUFU* after granule neuron specification (using the Math1-Cre line) does not lead to these defects (Jiwani et al., 2020). Thus, *SUFU*-associated infantile MB^{SHH} is a likely consequence of defects stemming from early stages of granule neuron lineage specification at embryonic stages.

Unfortunately, tumors initiated at embryonic stages are typically undetectable until several months after birth when tumorigenesis has significantly progressed. Thus, therapeutics for infantile MB must successfully curtail tumorigenic mechanisms at postnatal stages and minimally affect normal GNPs elsewhere in the cerebellum. Towards this goal, inhibiting localized FGF5 and FGF signaling activity may provide new paths toward the design of targeted treatments. Since we confirmed the occurrence of high FGF5 levels in a subset of infantile MB^{SHH} patients, FGF5 levels may be useful as a diagnostic biomarker for this patient population. This may predict the lack of efficacy of SHH-targeting compounds in curtailing tumor growth but could instead significantly impede cerebellar development. Importantly, detection of elevated FGF5 levels may identify patients who will be responsive to FGF-targeting treatments. Ultimately, we hope these studies facilitate the design of much needed precision medicines to address the distinct oncogenic mechanisms specifically and effectively in infantile MB^{SHH} patients while enabling normal progression of cerebellar development.

Acknowledgements

We thank Hirofumi Noguchi and other members of the Pleasure Lab for critical discussions, the UCSF Center for Advanced Light Microscopy for assistance with imaging, and William Krause for assistance with transcriptomics. Schematic diagrams were created with BioRender.com. This paper was typeset with the bioRxiv word template by @Chrelli: www.github.com/chrelli/bioRxiv-word-template. This work was supported by the NIH R01MH077694 and R01NS118995 (S.J.P.), R01MH077694-S1 (H.G.), NIH NIH/NCI K01CA201068 (O.R.Y.), and American Brain Tumor Association Grant #A131363 (O.R.Y.).

Competing interest statement

The authors declare no competing financial interest.

Materials and Methods

Animals

Mice carrying the floxed *Sufu* allele (*Sufu_{f/f}*) were kindly provided by Dr. Chi-Chung Hui (University of Toronto) and were genotyped as described elsewhere (Pospisilik et al., 2010). *hGFAP-cre* (Stock #004600) and *Gli1-LacZ* (Stock #008211) mice were obtained from Jackson Laboratories (Bar Harbor, ME, USA). Mice designated as controls did not carry the *Cre* transgene and may have either one of the following genotypes: *Sufu_{f/+}* or *Sufu_{f/f}*. All mouse lines were maintained in mixed strains, and analysis included male and female pups from each age group, although sex differences were not included in data reporting. All animal protocols were in accordance to the National Institute of Health regulations and approved by the UCSF Institutional Animal Care and Use Committee (IACUC).

In vivo treatment with FGFR inhibitor

The FGFR1-3 inhibitor, AZD4547 (Selleck Chemicals, #S2801) was dissolved sequentially in 4% Dimethyl Sulfoxide (DMSO), 30% Polyethylene glycol (PEG), 5% Tween-80, and water to make a 1mM solution. 1 μ l of 1mM AZD4547, or vehicle only as control, was injected into the lateral ventricle (~1 mm from the cerebellar midline) of pups for 5 days from P0/P1 using a 2.5 ml syringe (Model 62 RN; Hamilton Scientific). Pups remained with the mother until perfusion at P7 for analysis.

Immunohistochemistry and LacZ Staining

Perfusion, dissection, and immunofluorescence and LacZ staining were conducted according to standard protocols as previously described (Yabut et al., 2015). Briefly, P0/P1 brain tissues were fixed after dissection by direct immersion in 4% paraformaldehyde (PFA) and P7 and older postnatal brains fixed by intracardial perfusion followed by 2 h post-fixation. All fixed brains were cryoprotected a 15-30% sucrose gradient prior to cryosectioning. Cryostat sections were air dried and rinsed 3 $\times$ in PBS plus 0.2% Triton before blocking for 1 h in 10% normal lamb serum diluted in PBS with 0.2% Triton to prevent nonspecific binding. A heat-induced antigen retrieval protocol was performed on selective immunohistochemistry using 10 IzM Citric Acid at pH 6.0. Primary antibodies were diluted in 10% serum diluted in PBS with 0.2% Triton containing 4'-diamidino-2-phenylindole (DAPI); sections were incubated in primary antibody overnight at room temperature. The following antibodies were used: rabbit anti-Pax6 (1:250 dilution; Cat. #: 901301, Biolegend);

rabbit anti-NeuN (1:250 dilution; Cat. #: PA5-784-99, Invitrogen); mouse anti-Calretinin (1:250, Cat. #: AB5054, Millipore); rabbit anti-phospho-Erk1/2 (1:250 dilution; Cat. #: 4370, Cell Signaling); γ H2AX (1:100 dilution; Cat. #: 05-636, Millipore); mouse anti-Ki67 (1:100 dilution; Cat. #: 550809 BD Biosciences) and cleaved-Caspase 3 (1:250 dilution; Cat. #: 9661S, Cell Signaling). To detect primary antibodies, we used species-specific Alexa Fluor-conjugated secondary antibodies (1:500; Invitrogen) in 1X PBS-T for 1 h at room temperature, washed with 1X PBS, and coverslipped with Fluoromount-G (SouthernBiotech).

In Situ Hybridization

RNA Scope ISH was conducted for FGF15 and Ptch1. RNA Scope probes for Mm-FGF5 were designed commercially by the manufacturer (Advanced Cell Diagnostics, Inc.). RNA Scope Assay was performed using the RNA Scope Multiplex Fluorescent Reagent Kit V2 according to manufacturer's instructions. Detection of the probe was done with Opal 570 or Opal 520 reagent (Akoya Biosciences).

Western Blot Analysis

Western blot analyses were conducted according to standard protocols. Soluble extracts were loaded onto Criterion, 4-15% Tris-HCl 4 SDS-PAGE gels (Bio-Rad), separated at 120V, and transferred to PVDF membrane at 30V for 2 hours or overnight at 4°C. Membranes were blocked with 3% milk/1X TBS-T (Tris-buffered saline with 0.1% Tween 20) or 5% BSA/1X TBS-T for 1hr at room temperature, and incubated with primary antibodies diluted in blocking buffer overnight at 4°C, and secondary antibodies (1:5000 dilution; IR-Dye antibodies, LI-COR) for 1 hour at RT. Membranes were washed in 1X TBS-T and scanned using the Odyssey Infrared Imaging System (LI-COR). Primary antibodies were used as follows: rabbit anti-Gli1 (1:1000; Abcam); goat anti-Gli2 (1:1000; R&D Systems), rabbit anti-Gli3 (1:100; Santa Cruz), rabbit anti-Pax6 (1:1000 dilution; Cat. #: 901301, Biolegend); rabbit anti-NeuN (1:1000 dilution; Cat. #: PA5-784-99, Invitrogen); rabbit anti-GABA A Receptor α 6 (1:1000 dilution; Cat. #: PA5-77403, GABRA6; Invitrogen), and α -Tubulin (1:5000 dilution; Cat. #: ab4074, Abcam). Quantification and analysis were conducted using the Odyssey Image Studio Software (LI-COR). Protein levels were normalized to GAPDH protein levels. Levels of NeuN and GABRA6 were quantified in correlation with Pax6 levels (NeuN/Pax6 or GABRA6/Pax6) to determine the proportion of Pax6⁺ cells expressing mature granule neuron markers.

Image Analysis and Acquisition

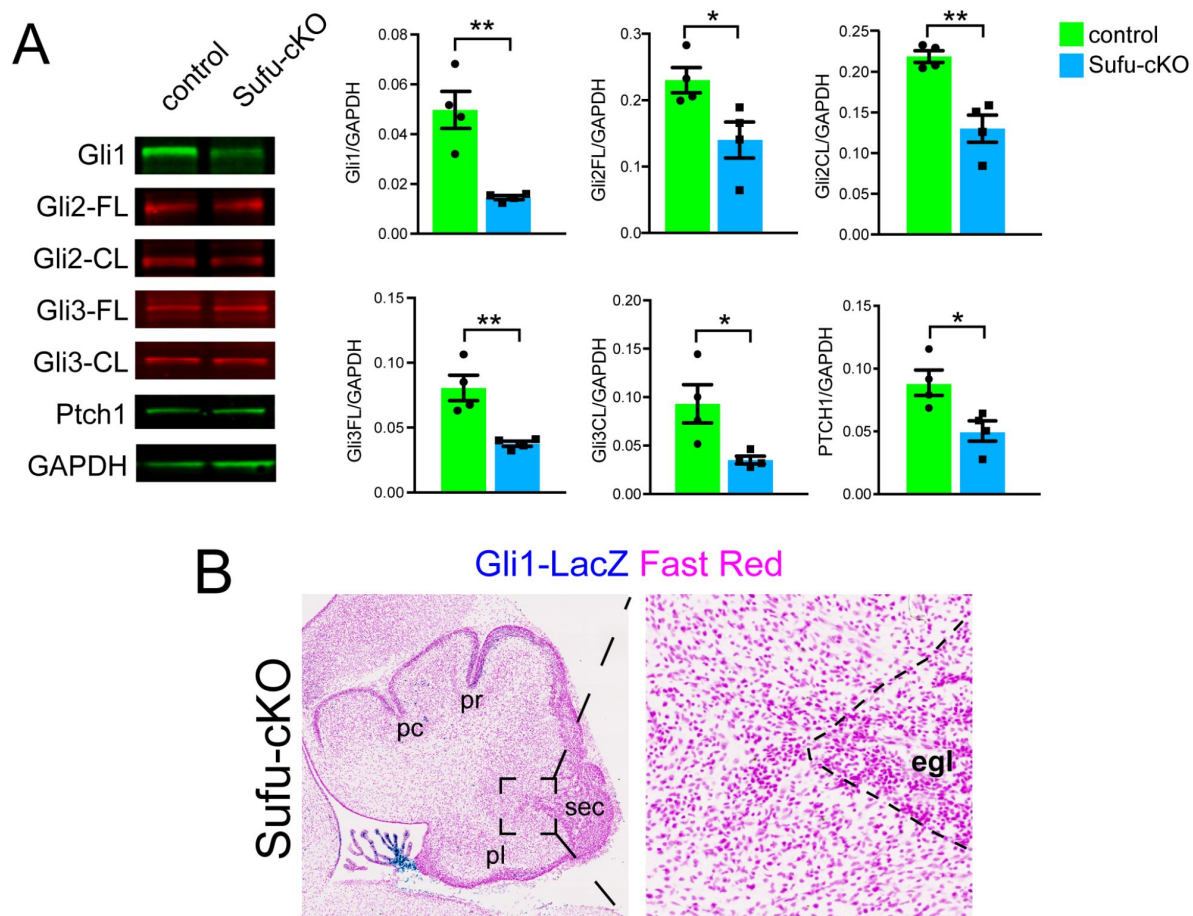
Images were acquired using a Nikon E600 microscope equipped with a QCapture Pro camera (QImaging), Zeiss AxioScan Z.1 (Zeiss, Thornwood, NY, USA) using the Zen 2 blue edition software (Zeiss, Thornwood, NY, USA), or the Nikon Ti inverted microscope with CSU-W1 large field of view confocal and Andor Zyla 4.2 sCMOS camera. All images were imported in tiff or jpeg format. Brightness, contrast, and background were adjusted equally for the entire image between controls and mutant using the "Brightness/Contrast" and "Levels" function from "Image/Adjustment" options in Adobe Photoshop or NIH ImageJ without any further modification. NIH Image J was used to threshold background levels between controls and mutant tissues to quantify fluorescence labeling. To quantify cell density, positively labeled cells within defined EGL regions, as defined in **Fig. 2A**, were counted. Measurements were obtained and averaged from 2-3 20 lzm thick and histologically matched sections encompassing the cerebellar midline (vermis).

Human MB Gene Expression

Expression values of FGF5 (ENSG00000138675) were obtained using Geo2R(Barrett et al., 2013) from published human MB subtype expression dataset GEO Accession no. GSE85217 (Cavalli et al., 2017). Analyses were conducted according to reported clinical data, subgroup and subtype classifications accompanying the dataset.

Statistics

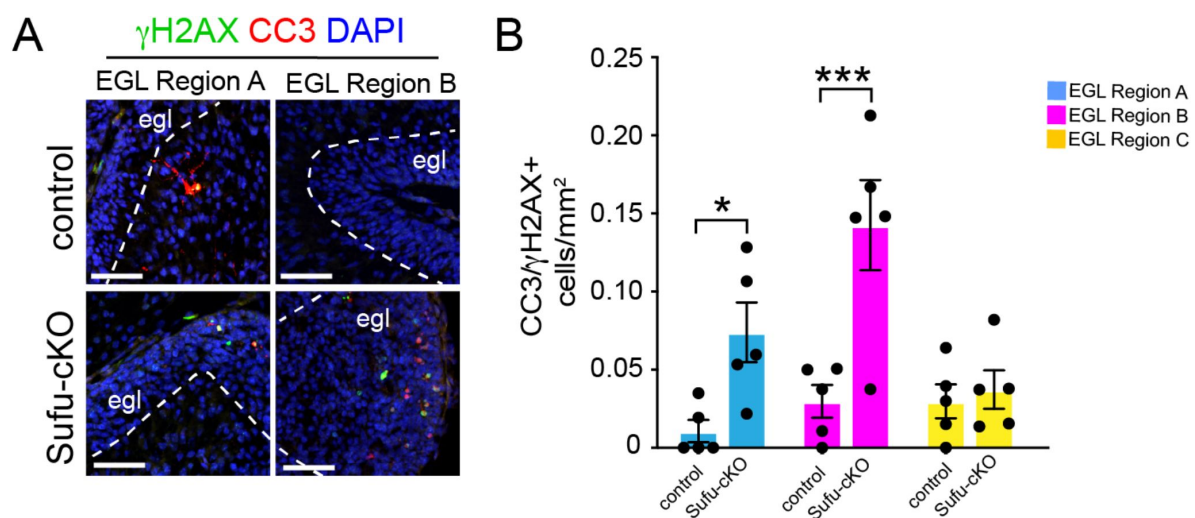
Prism 8.1 (GraphPad) was used for statistical analysis. Two sample experiments were analyzed by Student's *t* test and experiments with more than two parameters were analyzed by ANOVA. In 1- or 2-way ANOVA, when interactions were found, follow-up analyses were conducted for the relevant variables using Holm-Sidak's multiple comparisons test. All experiments were conducted at least in triplicate with sample sizes of $n = 3-6$ embryos/animals/slices per genotype. *P* value ≤ 0.05 was considered statistically significant. Graphs display the mean $\pm$ standard error of the mean (SEM).



Supplementary Figure 1

Reduced SHH signaling activity in the P0 Sufu-cKO cerebellum.

(A) Western blot analysis of cerebellar protein lysates from P0 control and Sufu-cKO mice showing significantly lower levels of total and cleaved versions of Gli transcription factors in the P0 Sufu-cKO cerebellum. * $p < 0.05$, ** $p < 0.01$ (B) b-galactosidase activity (blue), representing the Gli1-LacZ transgene, is largely absent in areas adjacent to the EGL along the secondary (sec) fissure of the P0 Sufu-cKO cerebellum.



Supplementary Figure 2

Evidence of pre-neoplastic lesions and high rates of cell death in Sufu-cKO granule neuron precursors.

(A) Double-immunofluorescence staining with (γ H2AX (green) and cleaved-caspase 3 (CC3; red), a marker for apoptotic cells, and DAPI labeling in regions A and B. Scale bars = 50 μ m **(B)** Quantification of the density of cells labeled with CC3 and (γ H2AX within each EGL regions. * p <0.05, *** p <0.001.

References

- Ahn S, Joyner AL (2005) **In vivo analysis of quiescent adult neural stem cells responding to Sonic hedgehog** *Nature* **437**:894–897 <https://doi.org/10.1038/nature03994>
- Allerstorfer S *et al.* (2008) **FGF5 as an oncogenic factor in human glioblastoma multiforme: autocrine and paracrine activities** *Oncogene* **27**:4180–4190 <https://doi.org/10.1038/onc.2008.61>
- Barrett T *et al.* (2013) **NCBI GEO: archive for functional genomics data sets--update** *Nucleic Acids Res* **41**:D991–5 <https://doi.org/10.1093/nar/gks1193>
- Cavalli FMG *et al.* (2017) **Intertumoral Heterogeneity within Medulloblastoma Subgroups** *Cancer Cell* **31**:737–754 <https://doi.org/10.1016/j.ccell.2017.05.005>
- Chen M-H, Wilson CW, Li Y-J, Lo Law KK, Lu C-S, Gacayan R, Zhang X, Hui C, Chuang P-T. (2009) **Cilium-independent regulation of Gli protein function by Sufu in Hedgehog signaling is evolutionarily conserved** *Genes Dev* **23**:1910–28 <https://doi.org/10.1101/gad.1794109>
- Clements DA, Wang JK, Dionne CA, Goldfarb M. (1993) **Activation of fibroblast growth factor (FGF) receptors by recombinant human FGF-5** *Oncogene* **8**:1311–6
- Corrales JD, Rocco GL, Blaess S, Guo Q, Joyner AL (2004) **Spatial pattern of sonic hedgehog signaling through Gli genes during cerebellum development** *Development* **131**:5581–90 <https://doi.org/10.1242/dev.01438>
- Emmenegger BA *et al.* (2013) **Distinct roles for fibroblast growth factor signaling in cerebellar development and medulloblastoma** *Oncogene* **32**:4181–4188 <https://doi.org/10.1038/onc.2012.440>
- Fogarty MP, Emmenegger BA, Gräsfeder LL, Oliver TG, Wechsler-Reya RJ, Lefkowitz RJ (2007) **Fibroblast growth factor blocks Sonic hedgehog signaling in neuronal precursors and tumor cells** *Proc Natl Acad Sci U S A*
- Frappart P-O, Lee Y, Russell HR, Chalhoub N, Wang Y-D, O'rii KE, Zhao J, Kondo N, Baker SJ, McKinnon PJ (2009) **Recurrent genomic alterations characterize medulloblastoma arising from DNA double-strand break repair deficiency** *Proc Natl Acad Sci U S A* **106**:1880–5 <https://doi.org/10.1073/pnas.0806882106>
- Gao Y, Ferguson DO, Xie W, Manis JP, Sekiguchi J, Frank KM, Chaudhuri J, Horner J, DePinho RA, Alt FW (2000) **Interplay of p53 and DNA-repair protein XRCC4 in tumorigenesis, genomic stability and development** *Nature* **404**:897–900 <https://doi.org/10.1038/35009138>
- Gudernova I *et al.* (2016) **Multikinase activity of fibroblast growth factor receptor (FGFR) inhibitors SU5402, PD173074, AZD1480, AZD4547 and BGJ398 compromises the use of small chemicals targeting FGFR catalytic activity for therapy of short-stature syndromes** *Hum Mol Genet* **25**:9–23 <https://doi.org/10.1093/hmg/ddv441>

- Guerrini-Rousseau L, Dufour C, Varlet P, Masliah-Planchon J, Bourdeaut F, Guillaud-Bataille M, Abbas R, Bertozzi A-I, Fouyssac F, et al. (2018) **Germline SUFU mutation carriers and medulloblastoma: clinical characteristics, cancer risk, and prognosis** *Neuro Oncol* **20**:1122–1132 <https://doi.org/10.1093/neuonc/nox228>
- Hattori Y, Miyake A, Mikami T, Ohta M, Itoh N. (1997) **Transient expression of FGF-5 mRNA in the rat cerebellar cortex during post-natal development** *Mol Brain Res* **47**:262–266 [https://doi.org/10.1016/S0169-328X\(97\)00065-X](https://doi.org/10.1016/S0169-328X(97)00065-X)
- Humke EW, Dorn K V., Milenkovic L, Scott MP, Rohatgi R. (2010) **The output of Hedgehog signaling is controlled by the dynamic association between Suppressor of Fused and the Gli proteins** *Genes Dev* **24**:670–82 <https://doi.org/10.1101/gad.1902910>
- Jiwani T, Kim JJ, Rosenblum ND (2020) **Suppressor of fused controls cerebellum granule cell proliferation by suppressing Fgf8 and spatially regulating Gli proteins** *Development* **147** <https://doi.org/10.1242/dev.170274>
- Kim JJ, Gill PS, Rotin L, van Eede M, Henkelman RM, Hui C-C, Rosenblum ND (2011) **Suppressor of fused controls mid-hindbrain patterning and cerebellar morphogenesis via GLI3 repressor** *J Neurosci* **31**:1825–36 <https://doi.org/10.1523/JNEUROSCI.2166-10.2011>
- Kim JJ, Jiwani T, Erwood S, Loree J, Rosenblum ND (2018) **Suppressor of fused controls cerebellar neuronal differentiation in a manner modulated by GLI3 repressor and Fgf15** *Dev Dyn* **247**:156–169 <https://doi.org/10.1002/dvdy.24526>
- Kong JH, Siebold C, Rohatgi R. (2019) **Biochemical mechanisms of vertebrate hedgehog signaling** *Development* **146** <https://doi.org/10.1242/dev.166892>
- Kool M *et al.* (2014) **Genome Sequencing of SHH Medulloblastoma Predicts Genotype-Related Response to Smoothed Inhibition** *Cancer Cell* **25**:393–405 <https://doi.org/10.1016/j.ccr.2014.02.004>
- Kool M, Korshunov A, Remke M, Jones DTW, Schlanstein M, Northcott PA, Cho Y-J, Koster J, Schouten-van Meeteren A, et al. (2012) **Molecular subgroups of medulloblastoma: an international meta-analysis of transcriptome, genetic aberrations, and clinical data of WNT, SHH, Group 3, and Group 4 medulloblastomas** *Acta Neuropathol* **123**:473–484 <https://doi.org/10.1007/s00401-012-0958-8>
- Lein ES *et al.* (2007) **Genome-wide atlas of gene expression in the adult mouse brain** *Nature* **445**:168–176 <https://doi.org/10.1038/nature05453>
- Leto K *et al.* (2016) **Consensus Paper: Cerebellar Development** *The Cerebellum* **15**:789–828 <https://doi.org/10.1007/s12311-015-0724-2>
- Mah L-J, El-Osta A, Karagiannis TC (2010) **γH2AX: a sensitive molecular marker of DNA damage and repair** *Leukemia* **24**:679–686 <https://doi.org/10.1038/leu.2010.6>
- Ornitz DM, Itoh N. (2015) **The Fibroblast Growth Factor signaling pathway** *Wiley Interdiscip Rev Dev Biol* **4**:215–66 <https://doi.org/10.1002/wdev.176>
- Ornitz DM, Xu J, Colvin JS, McEwen DG, MacArthur CA, Coulier F, Gao G, Goldfarb M. (1996) **Receptor specificity of the fibroblast growth factor family** *J Biol Chem* **271**:15292–7 <https://doi.org/10.1074/jbc.271.25.15292>

- Ostrom QT, Gittleman H, Xu J, Kromer C, Wolinsky Y, Kruchko C, Barnholtz-Sloan JS (2016) **CBTRUS Statistical Report: Primary Brain and Other Central Nervous System Tumors Diagnosed in the United States in 2009–2013** *Neuro Oncol* **18**:v1–v75 <https://doi.org/10.1093/neuonc/nov207>
- Ozawa K, Uruno T, Miyakawa K, Seo M, Imamura T. (1996) **Expression of the fibroblast growth factor family and their receptor family genes during mouse brain development** *Mol Brain Res* **41**:279–288 [https://doi.org/10.1016/0169-328X\(96\)00108-8](https://doi.org/10.1016/0169-328X(96)00108-8)
- Pazzaglia S, Tanori M, Mancuso M, Rebessi S, Leonardi S, Di Majo V, Covelli V, Atkinson MJ, Hahn H, Saran A. (2006) **Linking DNA damage to medulloblastoma tumorigenesis in patched heterozygous knockout mice** *Oncogene* **25**:1165–1173 <https://doi.org/10.1038/sj.onc.1209032>
- Pospisilik JA *et al.* (2010) **Drosophila genome-wide obesity screen reveals hedgehog as a determinant of brown versus white adipose cell fate** *Cell* **140**:148–60 <https://doi.org/10.1016/j.cell.2009.12.027>
- Remke M, Ramaswamy V. (2018) **Infant medulloblastoma — learning new lessons from old strata** *Nat Rev Clin Oncol* **15**:659–660 <https://doi.org/10.1038/s41571-018-0071-6>
- Schwalbe EC *et al.* (2017) **Novel molecular subgroups for clinical classification and outcome prediction in childhood medulloblastoma: A cohort study** *Lancet Oncol* **18**:958–971 [https://doi.org/10.1016/S1470-2045\(17\)30243-7](https://doi.org/10.1016/S1470-2045(17)30243-7)
- Smith KM, Williamson TL, Schwartz ML, Vaccarino FM (2012) **Impaired motor coordination and disrupted cerebellar architecture in Fgfr1 and Fgfr2 double knockout mice** *Brain Res* **1460**:12–24 <https://doi.org/10.1016/j.BRAINRES.2012.04.002>
- Tan I-L *et al.* (2018) **Lateral cerebellum is preferentially sensitive to high sonic hedgehog signaling and medulloblastoma formation** *Proc Natl Acad Sci U S A* **115**:3392–3397 <https://doi.org/10.1073/pnas.1717815115>
- Vanner RJ *et al.* (2014) **Quiescent Sox2+ Cells Drive Hierarchical Growth and Relapse in Sonic Hedgehog Subgroup Medulloblastoma** *Cancer Cell* **26**:33–47 <https://doi.org/10.1016/j.ccr.2014.05.005>
- Vladoiu MC *et al.* (2019) **Childhood cerebellar tumours mirror conserved fetal transcriptional programs** *Nature* **572**:67–73 <https://doi.org/10.1038/s41586-019-1158-7>
- Ward E, DeSantis C, Robbins A, Kohler B, Jemal A. (2014) **Childhood and adolescent cancer statistics, 2014** *CA Cancer J Clin* **64**:83–103 <https://doi.org/10.3322/caac.21219>
- Wechsler-Reya RJ, Scott MP (1999) **Control of Neuronal Precursor Proliferation in the Cerebellum by Sonic Hedgehog** *Neuron* **22**:103–114 [https://doi.org/10.1016/S0896-6273\(00\)80682-0](https://doi.org/10.1016/S0896-6273(00)80682-0)
- Wefers AK, Warmuth-Metz M, Pöschl J, von Bueren AO, Monoranu C-M, Seelos K, Peraud A, Tonn J-C, Koch A, *et al.* (2014) **Subgroup-specific localization of human medulloblastoma based on pre-operative MRI** *Acta Neuropathol* **127**:931–933 <https://doi.org/10.1007/s00401-014-1271-5>
- Yabut OR, Fernandez G, Huynh T, Yoon K, Pleasure SJ (2015) **Suppressor of Fused Is Critical for Maintenance of Neuronal Progenitor Identity during Corticogenesis** *Cell Rep* **12**:2021–2034 <https://doi.org/10.1016/j.celrep.2015.08.031>

- Yabut OR, Ng H-X, Yoon K, Arela JC, Ngo T, Pleasure SJ (2020) **The neocortical progenitor specification program is established through combined modulation of SHH and FGF signaling** *J Neurosci* <https://doi.org/10.1523/JNEUROSCI.2888-19.2020>
- Yaguchi Y, Yu T, Ahmed MU, Berry M, Mason I, Basson MA (2009) **Fibroblast growth factor (FGF) gene expression in the developing cerebellum suggests multiple roles for FGF signaling during cerebellar morphogenesis and development** *Dev Dyn* **238**:2058–2072 <https://doi.org/10.1002/dvdy.22013>
- Yin W-C *et al.* (2019) **Dual Regulatory Functions of SUFU and Targetome of GLI2 in SHH Subgroup Medulloblastoma** *Dev Cell* **48**:167–183 <https://doi.org/10.1016/j.devcel.2018.11.015>
- Zhuo L, Theis M, Alvarez-Maya I, Brenner M, Willecke K, Messing A. (2001) **hGFAP-cre transgenic mice for manipulation of glial and neuronal function in vivo** *Genesis* **31**:85–94

Editors

Reviewing Editor

Joshua Corbin

Children's National Hospital, Washington, United States of America

Senior Editor

Jonathan Cooper

Fred Hutchinson Cancer Research Center, Seattle, United States of America

Reviewer #1 (Public Review):

Summary:

SUFU modulates Sonic hedgehog (SHH) signaling and is frequently mutated in the B-subtype of SHH-driven medulloblastoma. The B-subtype occurs mostly in infants, is often metastatic, and lacks specific treatment. Yabut *et al.* found that *Fgf5* was highly expressed in the B-subtype of SHH-driven medulloblastoma by examining a published microarray expression dataset. They then investigated how *Fgf5* functions in the cerebellum of mice that have embryonic *Sufu* loss of function. This loss was induced using the hGFAP-cre transgene, which is expressed in multiple cell types in the developing cerebellum, including granule neuron precursors (GNPs) derived from the rhombic lip. By measuring the area of Pax6⁺ cells in the external granule cell layer (EGL) of *Sufu*-cKO mice at postnatal day 0, they find Pax6⁺ cells occupy a larger area in the posterior lobe adjacent to the secondary fissure, which is poorly defined. They show that *Fgf5* RNA and phosphoErk1/2 immunostaining are also higher in the same disrupted region. Some of the phosphoErk1/2⁺ cells are proliferative in the *Sufu*-cKO. Western blot analysis of Gli proteins that modulate SHH signaling found reduced expression and absence of Gli1 activity in the region of cerebellar dysgenesis in *Sufu*-cKO mice. This suggests the GNP expansion in this region is independent of SHH signaling. Amazingly, intraventricular injection of the FGFR1-2 antagonist AZD4547 from P0-4 and examined histologically at P7 found the treatment restored cytoarchitecture in the cerebella of *Sufu*-cKO mice. This is further supported by NeuN immunostaining in the internal granule cell layer, which labels mature, non-dividing neurons, and KI67 immunostaining, indicating dividing cells, and primarily found in the EGL. The mice were treated beginning at a timepoint when cerebellar cytoarchitecture was shown to be disrupted and it is indistinguishable from control following treatment. Figure 3 presents the most convincing and exciting data in this manuscript.

Sufu-cKO do not readily develop cerebellar tumors. The authors detected phosphorylated H2AX immunostaining, which labels double-strand breaks, in some cells in the EGL in regions of cerebellar dysgenesis in the Sufu-cKO, as was cleaved Caspase 3, a marker of apoptosis. P53, downstream of the double-strand break pathway, the protein was reduced in Sufu-cKO cerebellum. Genetically removing p53 from the Sufu-cKO cerebellum resulted in cerebellar tumors in 2-month old mice. The Sufu;p53-dKO cerebella at P0 lacked clear foliation, and the secondary fissure, even more so than the Sufu-cKO. Fgf5 RNA and signaling (pERK1/2) were also expressed ectopically.

The conclusions of the paper are largely supported by the data, but some data analysis need to be clarified and extended.

(1) The rationale for examining Fgf5 in medulloblastoma is not sufficiently convincing. The authors previously reported that Fgf15 was upregulated in neocortical progenitors of mice with conditional loss of Sufu (PMID: 32737167). In Figure 1, the authors report FGF5 expression is higher in SHH-type medulloblastoma, especially the beta and gamma subtypes mostly found in infants. These data were derived from a genome-wide dataset and are shown without correction for multiple testing, including other Fgfs. Showing the expression of other Fgfs with FDR correction would better substantiate their choice or moving this figure to later in the manuscript as support for their mouse investigations would be more convincing.

(2) The Sufu-cKO cerebellum lacks a clear anchor point at the secondary fissure and foliation is disrupted in the central and posterior lobes. It would be helpful for the authors to review Sudarov & Joyner (PMID: 18053187) for nomenclature specific to the developing cerebellum.

(3) The metrics used to quantify cerebellar perimeter and immunostaining are not sufficiently described. It is unclear whether the individual points in the bar graph represent a single section from independent mice, or multiple sections from the same mice. For example, in Figures 2B-D. This also applies to Figure 3C-D.

(4) The data on Fgf5 RNA expression presented in Figure 2E are not sufficiently convincing. The perimeter and cytoarchitecture of the cerebellum are difficult to see and the higher magnification shown in 2F should be indicated in 2E.

(5) The data presented in Figure 3 are not sufficiently convincing. The number of cells double positive for pErk and KI67 (Figure 3B) are difficult to see and appear to be few, suggesting the quantification may be unreliable.

(6) The data presented in Figure 4F-J would be more convincing with quantification. The Sufu;p53-dKO appears to have a thickened EGL across the entire vermis perimeter, and very little foliation, relative to control and single cKO cerebella. This is a more widespread effect than the more localized foliation disruption in the Sufu-cKO.

(7) Figure 5 does not convincingly summarize the results. Blue and purple cells in sagittal cartoon are not defined. Which cells express Fgf5 (or other Fgfs) has not been determined. The yellow cells are not defined in relation to the initial cartoon on the left.

<https://doi.org/10.7554/eLife.100767.1.sa3>

Reviewer #2 (Public Review):

Summary:

Mutations in SUFU are implicated in SHH medulloblastoma (MB). SUFU modulates Shh signaling in a context-dependent manner, making its role in MB pathology complex and not fully understood. This study reports that elevated FGF5 levels are associated with a specific

subtype of SHH MB, particularly in pediatric cases. The authors demonstrate that Sufu deletion in a mouse model leads to abnormal proliferation of granule cell precursors (GCPs) at the secondary fissure (region B), correlating with increased *Fgf5* expression. Notably, pharmacological inhibition of FGFR restores normal cerebellar development in Sufu mutant mice.

Strengths:

The identification of increased FGF5 in subsets of MB is novel and a key strength of the paper.

Weaknesses:

The study appears incomplete despite the potential significance of these findings. The current paper does not fully establish the causal relationship between *Fgf5* and abnormal cerebellar development, nor does it clarify its connection to SUFU-related MB. Some conclusions seem overstated, and the central question of whether FGFR inhibition can prevent tumor formation remains untested.

<https://doi.org/10.7554/eLife.100767.1.sa2>

Reviewer #3 (Public Review):

Summary:

The interaction between FGF signaling and SHH-mediated GNP expansion in MB, particularly in the context of Sufu LoF, has just begun to be understood. The manuscript by Yabut et al. establishes a connection between ectopic FGF5 expression and GNP over-expansion in a late-stage embryonic Sufu LoF model. The data provided links region-specific interaction between aberrant FGF5 signaling with the SHH subtype of medulloblastoma. New data from Yabut et al. suggest that ectopic FGF5 expression correlates with GNP expansion near the secondary fissure in Sufu LoF cerebella. Furthermore, pharmacological blockade of FGF signaling inhibits GNP proliferation. Interestingly, the data indicate that the timing of conditional Sufu deletion (E13.5 using the hGFAP-Cre line) results in different outcomes compared to later deletion (using Math1-cre line, Jiواني et al., 2020). This study provides significant insights into the molecular mechanisms driving GNP expansion in SHH subgroup MB, particularly in the context of Sufu LoF. It highlights the potential of targeting FGF5 signaling as a therapeutic strategy. Additionally, the research offers a model for better understanding MB subtypes and developing targeted treatments.

Strengths:

One notable strength of this study is the extraction and analysis of ectopic FGF5 expression from a subset of MB patient tumor samples. This translational aspect of the study enhances its relevance to human disease. By correlating findings from mouse models with patient data, the authors strengthen the validity of their conclusions and highlight the potential clinical implications of targeting FGF5 in MB therapy.

The data convincingly show that FGFR signaling activation drives GNP proliferation in Sufu, conditional knockout models. This finding is supported by robust experimental evidence, including pharmacological blockade of FGF signaling, which effectively inhibits GNP proliferation. The clear demonstration of a functional link between FGFR signaling and GNP expansion underscores the potential of FGFR as a therapeutic target in SHH subgroup medulloblastoma.

Previous studies have demonstrated the inhibitory effect of FGF2 on tumor cell proliferation in certain MB types, such as the ptc mutant (Fogarty et al., 2006)(Yaguchi et al., 2009).

Findings in this manuscript provide additional support suggesting multiple roles for FGF signaling in cerebellar patterning and development.

Weaknesses:

In the GEO dataset analysis, where FGF5 expression is extracted, the reporting of the P-value lacks detail on the statistical methods used, such as whether an ANOVA or t-test was employed. Providing comprehensive statistical methodologies is crucial for assessing the rigor and reproducibility of the results. The absence of this information raises concerns about the robustness of the statistical analysis.

Another concern is related to the controls used in the study. Cre recombinase induces double-strand DNA breaks within the loxP sites, and the control mice did not carry the Cre transgene (as stated in the Method section), while Sufu-cKO mice did. This discrepancy necessitates an additional control group to evaluate the effects of Cre-induced double-strand breaks on phosphorylated H2AX-DSB signaling. Including this control would strengthen the validity of the findings by ensuring that observed effects are not artifacts of Cre recombinase activity.

Although the use of the hGFAP-Cre line allows genetic access to the late embryonic stage, this also targets multiple celltypes, including both GNPs and cerebellar glial cells. However, the authors focus primarily on GNPs without fully addressing the potential contributions of neuron-glial interaction. This oversight could limit the understanding of the broader cellular context in which FGF signaling influences tumor development.

<https://doi.org/10.7554/eLife.100767.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public Review):

Summary:

SUFU modulates Sonic hedgehog (SHH) signaling and is frequently mutated in the B-subtype of SHH-driven medulloblastoma. The B-subtype occurs mostly in infants, is often metastatic, and lacks specific treatment. Yabut et al. found that Fgf5 was highly expressed in the B-subtype of SHH-driven medulloblastoma by examining a published microarray expression dataset. They then investigated how Fgf5 functions in the cerebellum of mice that have embryonic Sufu loss of function. This loss was induced using the hGFAP-cre transgene, which is expressed in multiple cell types in the developing cerebellum, including granule neuron precursors (GNPs) derived from the rhombic lip. By measuring the area of Pax6+ cells in the external granule cell layer (EGL) of Sufu-cKO mice at postnatal day 0, they find Pax6+ cells occupy a larger area in the posterior lobe adjacent to the secondary fissure, which is poorly defined. They show that Fgf5 RNA and phosphoErk1/2 immunostaining are also higher in the same disrupted region. Some of the phosphoErk1/2+ cells are proliferative in the Sufu-cKO. Western blot analysis of Gli proteins that modulate SHH signaling found reduced expression and absence of Gli1 activity in the region of cerebellar dysgenesis in Sufu-cKO mice. This suggests the GNP expansion in this region is independent of SHH signaling. Amazingly, intraventricular injection of the FGFR1-2 antagonist AZD4547 from P0-4 and examined histologically at P7 found the treatment restored cytoarchitecture in the cerebella of Sufu-cKO mice. This is further supported by NeuN immunostaining in the internal granule cell layer, which labels mature, non-dividing neurons, and KI67 immunostaining, indicating dividing cells, and primarily found in the EGL. The mice were treated beginning at a timepoint when

cerebellar cytoarchitecture was shown to be disrupted and it is indistinguishable from control following treatment. Figure 3 presents the most convincing and exciting data in this manuscript.

Sufu-cKO do not readily develop cerebellar tumors. The authors detected phosphorylated H2AX immunostaining, which labels double-strand breaks, in some cells in the EGL in regions of cerebellar dysgenesis in the Sufu-cKO, as was cleaved Caspase 3, a marker of apoptosis. P53, downstream of the double-strand break pathway, the protein was reduced in Sufu-cKO cerebellum. Genetically removing p53 from the Sufu-cKO cerebellum resulted in cerebellar tumors in 2-month old mice. The Sufu;p53-dKO cerebella at P0 lacked clear foliation, and the secondary fissure, even more so than the Sufu-cKO. Fgf5 RNA and signaling (pERK1/2) were also expressed ectopically.

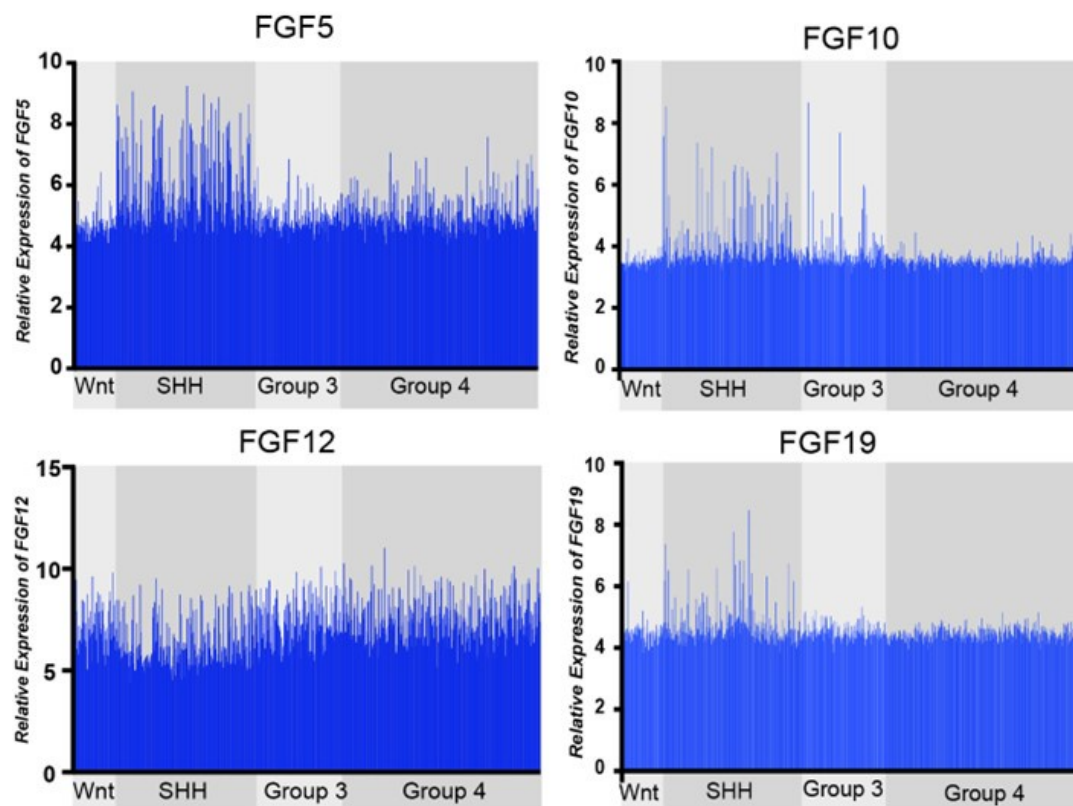
The conclusions of the paper are largely supported by the data, but some data analysis need to be clarified and extended.

(1) The rationale for examining Fgf5 in medulloblastoma is not sufficiently convincing. The authors previously reported that Fgf15 was upregulated in neocortical progenitors of mice with conditional loss of Sufu (PMID: 32737167). In Figure 1, the authors report FGF5 expression is higher in SHH-type medulloblastoma, especially the beta and gamma subtypes mostly found in infants. These data were derived from a genome-wide dataset and are shown without correction for multiple testing, including other Fgfs. Showing the expression of other Fgfs with FDR correction would better substantiate their choice or moving this figure to later in the manuscript as support for their mouse investigations would be more convincing.

To assess FGF5 (ENSG00000138675) expression in MB tissues, we used Geo2R (Barrett et al., 2013) to analyze published human MB subtype expression arrays from accession no. GSE85217 (Cavalli et al., 2017). GEO2R is an interactive web tool that compares expression levels of genes of interest (GOI) between sample groups in the GEO series using original submitter-supplied processed data tables. We entered the GOI Ensembl ID and organized data sets according to age and MB subgroup or MBSHH subtype classifications. GEO2R results presented gene expression levels as a table ordered by FDR-adjusted (Benjamini & Hochberg) p-values, with significance level cut-off at 0.05, processed by GEO2R's built-in limma statistical test. Resulting data were subsequently exported into Prism (GraphPad). We generated scatter plots presenting FGF5 expression levels across all MB subgroups (Figure 1A) and MBSHH subtypes (Figure 1D). We performed additional statistical analyses to compare FGF5 expression levels between MB subgroups and MBSHH subtypes and graphed these data as violin plots (Figure 1B, 1C, and 1E). For these analyses, we used one-way ANOVA with Holm-Sidak's multiple comparisons test, single pooled variance. *P* value ≤ 0.05 was considered statistically significant. Graphs display the mean $\pm$ standard error of the mean (SEM).

Author response image 1.

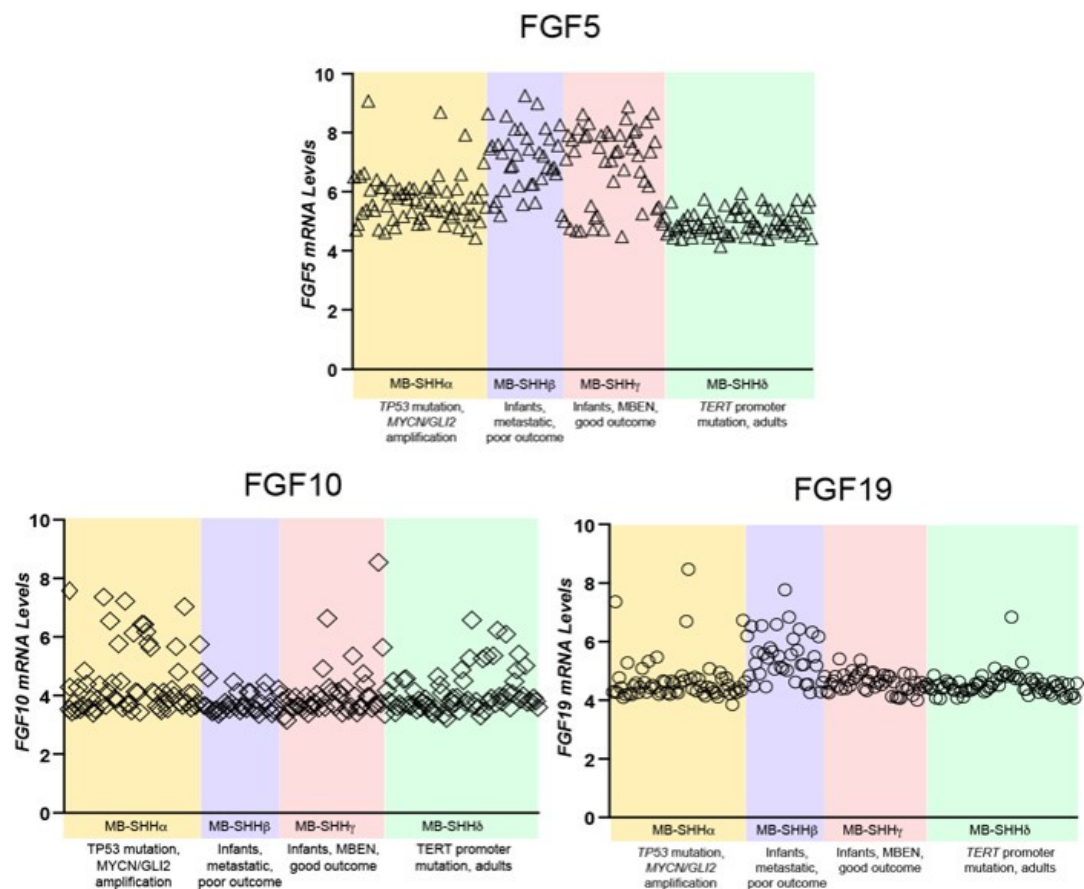
Comparative expression of FGF ligands, FGF5, FGF10, FGF12, and FGF19, across all MB subgroups. FGF12 expression is not significantly different, while FGF5, FGF10, and FGF19, show distinct upregulation in MBSHH subgroup (MBWNT n=70, MBSHH n=224, MBGR3 n=143, MBGR4 n=326).



Expression of the 21 known FGF ligands were also analyzed. Many FGFs did not exhibit differential expression levels in MBSHH compared to other MB subgroups, such as with FGF12 in Figure 1. FGF5, FGF10, and FGF19 (the human orthologue of mouse FGF15) all showed specific upregulation in MBSHH compared to other MB subgroups (Author response image 1), supporting our previous observations that FGF15 is a downstream target of SHH signaling (Yabut et al., 2020), as the reviewer pointed out. However, further stratification of MBSHH patient data revealed that only FGF5 specifically showed upregulation in infants with MBSHH (MBSHHb and MBSHHg Author response image 2) indicating a more prominent role for FGF5 in the developing cerebellum and driver of MBSHH tumorigenesis in this dynamic environment.

Author response image 2.

Comparative expression of FGF5, FGF10, and FGF19 in different MBSHH subtypes. FGF5 specifically show mRNA relative levels above 6 in 81% of MBSHH infant patient tumors (n=80 MBSHHb and MBSHHg tumors) unlike 35% of MBSHHa (n=65) or 0% of MBSHHd (n=75) tumors.



(2) The *Sufu*-cKO cerebellum lacks a clear anchor point at the secondary fissure and foliation is disrupted in the central and posterior lobes. It would be helpful for the authors to review Sudarov & Joyner (PMID: 18053187) for nomenclature specific to the developing cerebellum.

The reviewers are correct that the cerebellar foliation is severely disrupted in central and posterior lobes, as per Sudarov and Joyner (Neural Development 2007). This nomenclature may be referred to describe the regions referred in this manuscript.

(3) The metrics used to quantify cerebellar perimeter and immunostaining are not sufficiently described. It is unclear whether the individual points in the bar graph represent a single section from independent mice, or multiple sections from the same mice. For example, in Figures 2B-D. This also applies to Figure 3C-D.

All quantification were performed from 2-3 20 μ m cerebellar sections of 3-6 independent mice per genotype analyzed. Individual points in the bar graphs represent the average cell number (quantified from 2-3 sections) from each mice. Figure 2B show data points from n=4 mice per genotype. Figure 2C show data from n=3 mice per genotype. Figure 2D show data from n=6 mice per genotype. Figure 3C-D show data from n=3 mice per genotype.

(4) The data on *Fgf5* RNA expression presented in Figure 2E are not sufficiently convincing. The perimeter and cytoarchitecture of the cerebellum are difficult to see and the higher magnification shown in 2F should be indicated in 2E.

The lack of foliation in *Sufu*-cKO cerebellum is clear particularly when visualizing the perimeter via DAPI labeling (Figure 2E). The expression area of *FGF5* is also visibly larger, given that all images in Figure 2E are presented in the same scale (scale bars = 500 μ m).

(5) The data presented in Figure 3 are not sufficiently convincing. The number of cells double positive for pErk and KI67 (Figure 3B) are difficult to see and appear to be few, suggesting the quantification may be unreliable.

We used KI67+ expression to provide a molecular marker of regions to be quantified in both WT and *Sufu*-cKO sections. Quantification of labeled cells were performed in images obtained by confocal microscopy, enabling imaging of 1-2 μ m optical slices since Ki67 or pERK expression might not localize within the same cellular compartments. We relied on continuous DAPI nuclear staining to distinguish individual cells in each optical slice and the colocalization of of Ki67 and pERK. All quantification were performed from 2-3 20 μ m cerebellar sections of 3-6 independent mice per genotype analyzed. Individual points in the bar graphs represent the average cell number (quantified from 2-3 sections) from each mice.

*(6) The data presented in Figure 4F-J would be more convincing with quantification. The *Sufu*;p53-dKO appears to have a thickened EGL across the entire vermis perimeter, and very little foliation, relative to control and single cKO cerebella. This is a more widespread effect than the more localized foliation disruption in the *Sufu*-cKO.*

We agree with the reviewers that quantification of these phenotypes provide a solid measure of the defects. The phenotypes of *Sufu*;p53-dKO cerebellum are so profound requiring in-depth characterization that will be the focus of future studies.

*(7) Figure 5 does not convincingly summarize the results. Blue and purple cells in sagittal cartoon are not defined. Which cells express *Fgf5* (or other *Fgfs*) has not been determined. The yellow cells are not defined in relation to the initial cartoon on the left.*

The revised manuscript will address this confusion by clearly labeling the cells and their roles in the schematic diagram.

Reviewer #2 (Public Review):

Summary:

*Mutations in *SUFU* are implicated in SHH medulloblastoma (MB). *SUFU* modulates *Shh* signaling in a context-dependent manner, making its role in MB pathology complex and not fully understood. This study reports that elevated *FGF5* levels are associated with a specific subtype of SHH MB, particularly in pediatric cases. The authors demonstrate that *Sufu* deletion in a mouse model leads to abnormal proliferation of granule cell precursors (GCPs) at the secondary fissure (region B), correlating with increased *Fgf5* expression. Notably, pharmacological inhibition of FGFR restores normal cerebellar development in *Sufu* mutant mice.*

Strengths:

*The identification of increased *FGF5* in subsets of MB is novel and a key strength of the paper.*

Weaknesses:

*The study appears incomplete despite the potential significance of these findings. The current paper does not fully establish the causal relationship between *Fgf5* and*

abnormal cerebellar development, nor does it clarify its connection to SUFU-related MB. Some conclusions seem overstated, and the central question of whether FGFR inhibition can prevent tumor formation remains untested.

Reviewer #3 (Public Review):

Summary:

The interaction between FGF signaling and SHH-mediated GNP expansion in MB, particularly in the context of Sufu LoF, has just begun to be understood. The manuscript by Yabut et al. establishes a connection between ectopic FGF5 expression and GNP over-expansion in a late-stage embryonic Sufu LoF model. The data provided links region-specific interaction between aberrant FGF5 signaling with the SHH subtype of medulloblastoma. New data from Yabut et al. suggest that ectopic FGF5 expression correlates with GNP expansion near the secondary fissure in Sufu LoF cerebella. Furthermore, pharmacological blockade of FGF signaling inhibits GNP proliferation. Interestingly, the data indicate that the timing of conditional Sufu deletion (E13.5 using the hGFAP-Cre line) results in different outcomes compared to later deletion (using Math1-cre line, Jiwani et al., 2020). This study provides significant insights into the molecular mechanisms driving GNP expansion in SHH subgroup MB, particularly in the context of Sufu LoF. It highlights the potential of targeting FGF5 signaling as a therapeutic strategy. Additionally, the research offers a model for better understanding MB subtypes and developing targeted treatments.

Strengths:

One notable strength of this study is the extraction and analysis of ectopic FGF5 expression from a subset of MB patient tumor samples. This translational aspect of the study enhances its relevance to human disease. By correlating findings from mouse models with patient data, the authors strengthen the validity of their conclusions and highlight the potential clinical implications of targeting FGF5 in MB therapy.

The data convincingly show that FGFR signaling activation drives GNP proliferation in Sufu, conditional knockout models. This finding is supported by robust experimental evidence, including pharmacological blockade of FGF signaling, which effectively inhibits GNP proliferation. The clear demonstration of a functional link between FGFR signaling and GNP expansion underscores the potential of FGFR as a therapeutic target in SHH subgroup medulloblastoma.

Previous studies have demonstrated the inhibitory effect of FGF2 on tumor cell proliferation in certain MB types, such as the ptc mutant (Fogarty et al., 2006)(Yaguchi et al., 2009). Findings in this manuscript provide additional support suggesting multiple roles for FGF signaling in cerebellar patterning and development.

Weaknesses:

In the GEO dataset analysis, where FGF5 expression is extracted, the reporting of the P-value lacks detail on the statistical methods used, such as whether an ANOVA or t-test was employed. Providing comprehensive statistical methodologies is crucial for assessing the rigor and reproducibility of the results. The absence of this information raises concerns about the robustness of the statistical analysis.

The revised manuscript will include the following detailed explanation of the statistical analyses of the GEO dataset:

For the analysis of expression values of FGF5 (ENSG00000138675), we obtained these values using Geo2R (Barrett et al., 2013), which directly analyze published human MB subtype

expression arrays from accession no. GSE85217 (Cavalli et al., 2017). GEO2R is an interactive web tool that compares expression levels of genes of interest (GOI) between sample groups in the GEO series using original submitter-supplied processed data tables. We simply entered the GOI Ensembl ID and organized data sets according to age and MB subgroup or MBSHH subtype classifications. GEO2R results presented gene expression levels as a table ordered by FDR-adjusted (Benjamini & Hochberg) p-values, with significance level cut-off at 0.05, processed by GEO2R's built-in limma statistical test. Resulting data were subsequently exported into Prism (GraphPad). We generated scatter plots presenting FGF5 expression levels across all MB subgroups (Figure 1A) and MBSHH subtypes (Figure 1D). We performed additional statistical analyses to compare FGF5 expression levels between MB subgroups and MBSHH subtypes and graphed these data as violin plots (Figure 1B, 1C, and 1E). For these analyses, we used one-way ANOVA with Holm-Sidak's multiple comparisons test, single pooled variance. *P* value ≤ 0.05 was considered statistically significant. Graphs display the mean $\pm$ standard error of the mean (SEM). Sample sizes were:

Author response table 1.

MB ^{WNT} n=70	MB ^{SHHα} n=66
MB ^{SHH} n=224	MB ^{SHHβ} n=35
MB ^{GR3} n=143	MB ^{SHHγ} n=47
MB ^{GR4} n=326	MB ^{SHHδ} n=76

Another concern is related to the controls used in the study. Cre recombinase induces double-strand DNA breaks within the loxP sites, and the control mice did not carry the Cre transgene (as stated in the Method section), while Sufu-cKO mice did. This discrepancy necessitates an additional control group to evaluate the effects of Cre-induced double-strand breaks on phosphorylated H2AX-DSB signaling. Including this control would strengthen the validity of the findings by ensuring that observed effects are not artifacts of Cre recombinase activity.

The breeding scheme we used to generate homozygous SUFU conditional mutants will not generate pups carrying only hGFAP-Cre. Thus, we are unable to compare expression of gH2AX expression in littermates that do not carry loxP sites. The reviewer is correct in pointing out the possibility of Cre recombinase activity inducing double-strand breaks on its own. However, it is likely that any hGFAP-Cre induced double-strand breaks does not sufficiently cause the phenotypes we observed in homozygous mutants (Sufu-cKO) mice because the cerebellum of mice carry heterozygous SUFU mutations (hGFAP-Cre;Sufu-fl/+) do not display the profound cerebellar phenotypes observed in Sufu-cKO mice. We cannot rule out, however, any undetectable abnormalities that could be present which may require further analyses.

Although the use of the hGFAP-Cre line allows genetic access to the late embryonic stage, this also targets multiple celltypes, including both GNPs and cerebellar glial cells. However, the authors focus primarily on GNPs without fully addressing the potential contributions of neuron-glial interaction. This oversight could limit the understanding of the broader cellular context in which FGF signaling influences tumor development.

The reviewer is correct in that hGFAP-Cre also targets other cell types, such as cerebellar glial cells, which are generated when Cre-expression has begun. It is possible that cerebellar glial cell development is also compromised in Sufu-cKO mice and may disrupt neuron-glial interaction, due to or independently of FGF signaling. In-depth studies are required to

interrogate how loss of SUFU specifically affect development of cerebellar glial cells and influence their cellular interactions in the developing cerebellum.

<https://doi.org/10.7554/eLife.100767.1.sa0>